AMIA 2026 Annual Symposium

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Causal AI for Health: Moving from Prediction to Actionable Inference

PART II WORKSHOP | AMIA 2026 KDDM Working Group Collaborative Workshop

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HEALTH
UNIVERSITY OF UTAH

Learning Objectives

At the conclusion of this presentation the learner will be able to:

1. Distinguish causal inference from predictive modeling and determine when causal methods are required for clinical or policy decisions
2. Identify key sources of bias and confounding in observational health data and select appropriate study designs, such as target trial emulation, for estimating treatment effects
3. **Describe emerging approaches that integrate structural causal models and generative AI** with machine learning to improve fairness, transportability, and real-world deployment
    - Survey learners – “Choose your use case!
    - Hands-on with GenAI Tools (as time permits): ChatGPT, MCP, Figma

What participants will do

Throughout this section we will:

1. Survey the learners – “Choose your use case!
2. Hands-on with GenAI Tools (as time permits): ChatGPT, MCP, Figma
3. Build a clinician-governed protype using causal and mechanistic reasoning
4. Apply safe implementation guidelines and stress-testing output
5. Q& A

Pick Your Use Case | 3 Option Poll

QR Code here

1. **Etiologic modeling**
   - Explain why an outcome may have occurred and identify potentially modifiable contributors.
2. **Mechanism-based differential reasoning**
   - Differentiate possible explanations for an uncertain clinical presentation.
3. **Interventional decision-support**
   - Assess a clinical workflow intervention's benefits and tradeoffs.

1- Etiologic Modeling Assistant

Core question

“What may have contributed to an observed clinical, quality, or workflow outcome, through what plausible mechanisms, and what evidence would be needed before a team treats that explanation as credible?”

2 - Mechanism Based Differential Reasoning

Core question

*"What mechanisms could **explain an uncertain clinical presentation,** what makes each explanation more or **less plausible**, what **remains urgent to exclude**, and what additional information would most meaningfully **distinguish among alternatives**?"*

3 - Interventional Decision Support

Core question

*“Could a proposed clinical workflow intervention plausibly **improve the intended outcome**, through what mechanism, for whom, under **what conditions**, and with **what tradeoffs**?”*

Safe Implementation Guidelines

- Follow institutional guidelines
- Require the model to separate observed facts, assumptions, plausible mechanisms, unknowns, and next evidence needs.
- Do not allow the prototype to fabricate information, present a definitive diagnosis, prescribe treatment, or convert output directly into clinical action.
- For instability or high-risk uncertainty, state that local escalation and clinician assessment take priority.
- Do not claim regulatory compliance. Identify validation, governance, privacy, security, and oversight as organization-specific requirements.

Let’s Get Ready!

- Get resources: QR Code
  - Documents & Files

QR Code here

- Login to ChatGPT
- we will be working in:
  - Work mode
  - Skills
  - Projects
  - Visualization

- Example: Output With & Without Skills

video

ChatGPT - Build Own Skills/ Prototype

- Follow instructions in handout
- Discuss prompt design
- Use repository to access premade skills

• Frame the causal inquiry

Worksheet field	Prompt for participants	Example
Outcome	What happened that we want to explain?	Delayed escalation after signs of deterioration
Setting and population	Where and for whom?	Adult medical-surgical inpatients
Unit of analysis	What is being examined?	Care episode and associated workflow
Intended user	Who will use the analysis?	Quality, safety, informatics, or clinical-review team
Intended use	What will the output support?	Hypothesis generation and evidence gathering, not blame assignment

Stress-Testing Output

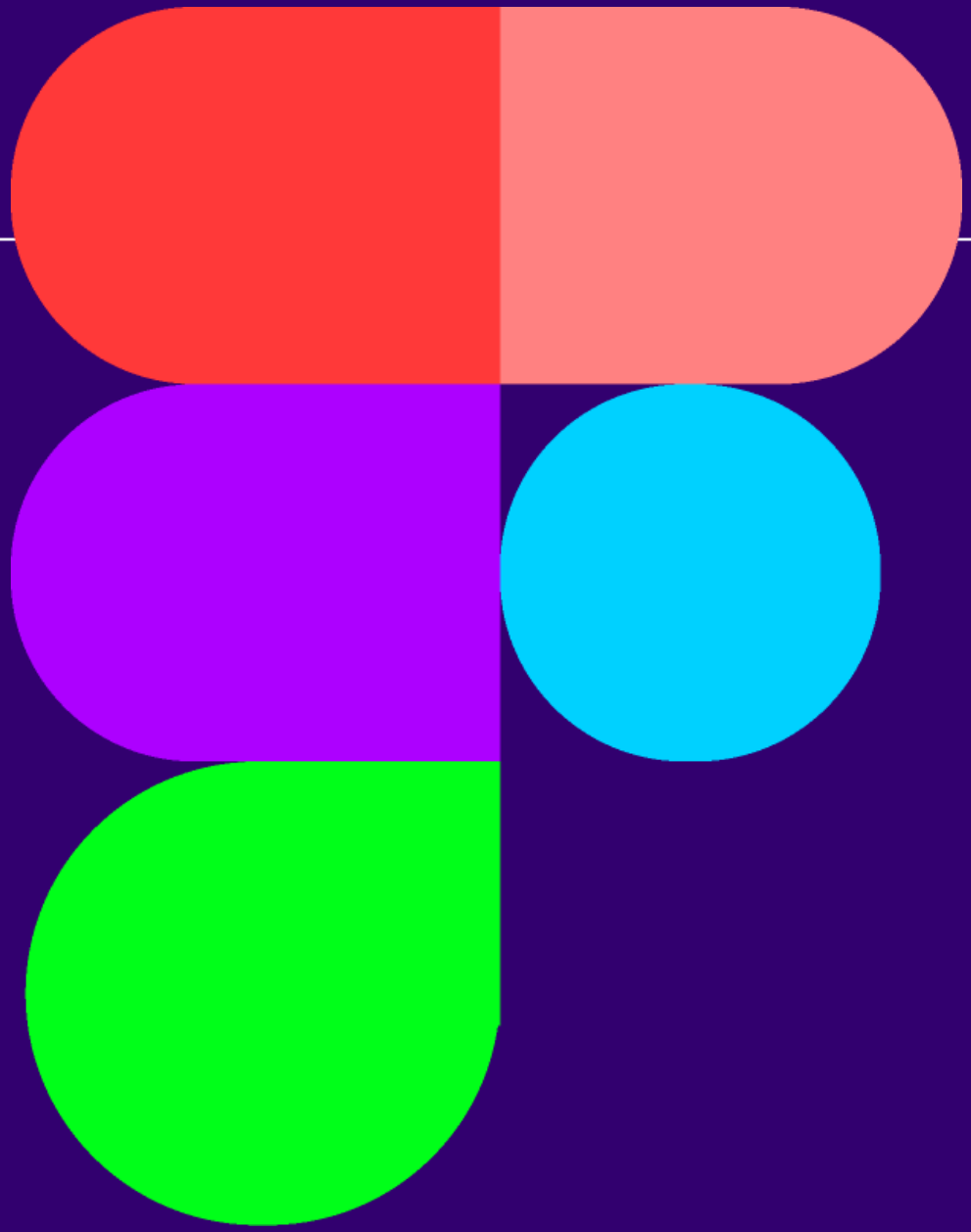
- Test at least 3 times
  - Something it should know
  - Something it should not know
  - Extreme cases
- Reiterate/ edit prompt instructions
- Red team

Using a Repository

- Accessing additional skills through a repository
- Video Demo

Projects with Figma

- What is Figma?
- Applying Figma
- Example Project



Projects using MCP

- Video Demo

Questions?